Maschke's Complete Reducibility Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

An arbitrary projection need not respect the group action. Averaging its conjugates creates a projection that does. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RT-01 / Representation Theory and Lie Theory / Report v1

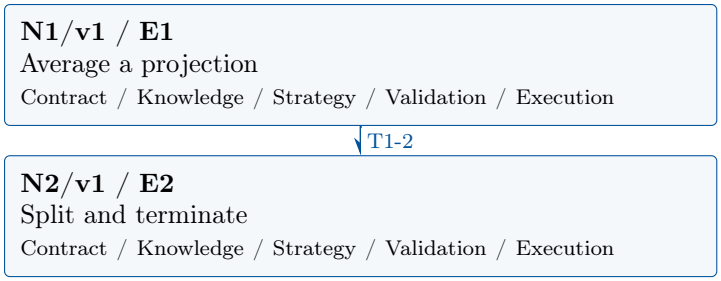
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let G be a finite group, k a field in which $|G|$ is invertible, and V a finite-dimensional k-representation. Every invariant subspace W has an invariant complement, and V is a direct sum of irreducible representations. A representation is a homomorphism $\rho : G \to \mathrm{GL}(V)$; irreducible means nonzero with no proper nonzero invariant subspace.*

A. Foundations and permitted imports

Use finite-dimensional basis extension to choose a linear projection onto a subspace, invariance under reindexing a finite group sum, and induction on dimension. The field may have positive characteristic provided it does not divide $|G|$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $P = |G|^{-1} \sum_g \rho(g) p \rho(g)^{-1}$ is equivariant, has image W, and restricts to the identity on W.

Receiving activity. N2.1

Compatibility. E1/v1 and finite $\dim V$. Check: The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Average a projection

Object and role. Average $\rho(g)p\rho(g)^{-1}$ to enforce equivariance without losing the restriction to W. **Local obligation:** $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image W, and restricts to the identity on W.

Inputs. An invariant $W \subset V$ and any linear projection $p : V \to W$ with $p|_W = \mathrm{id}$.

C — Contract

Publish E1/v1 only after establishing this obligation: $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image W, and restricts to the identity on W.

K — Knowledge

An invariant $W \subset V$ and any linear projection $p : V \to W$ with $p|_W = \mathrm{id}$. Retain the stated assumptions and import versions.

S — Strategy

Average $\rho(g)p\rho(g)^{-1}$ to enforce equivariance without losing the restriction to W.

V — Validation

Division by $|G|$ must be legal, and each conjugate must still land in W.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Choose a vector-space complement by extending a basis of W, and let p be its projection. Each conjugate of p lands in W, since W is invariant under every $\rho(g)$. On $w \in W$, that conjugate sends w to itself. Thus the average P also maps into W and is the identity there.

N1.2 For $h \in G$, conjugating P by $\rho(h)$ replaces the index g by hg, a permutation of G. Therefore P commutes with $\rho(h)$. Its restriction to its image is the identity, so $P^2 = P$ and $\operatorname{im} P = W$.

Export E1/v1. $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image W, and restricts to the identity on W.

Acceptance gate. Division by $|G|$ must be legal, and each conjugate must still land in W.

Targeted repair. If p is assumed equivariant from basis extension alone, replace it by the conjugation average and verify the reindexing. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Split and terminate

Object and role. Use the equivariant kernel as a complement, then induct on dimension. **Local obligation:** $V = W \oplus \ker P$ as representations, and every such V is a finite direct sum of irreducibles.

Inputs. E1/v1 and finite $\dim V$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $V = W \oplus \ker P$ as representations, and every such V is a finite direct sum of irreducibles.

K — Knowledge

E1/v1 and finite $\dim V$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Use the equivariant kernel as a complement, then induct on dimension.

V — Validation

The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Every v decomposes as $Pv+(v-Pv)$, with the first term in W and the second in $\ker P$. Their intersection is zero because $P|_W$ is the identity. If $u \in \ker P$, then $P\rho(g)u = \rho(g)Pu = 0$, so the complement is invariant.

N2.2 For $\dim V = 0$ the decomposition is the empty sum. A nonzero irreducible V is already one summand. Otherwise choose a proper nonzero invariant W, split it as above, and apply the induction hypothesis to both summands of strictly smaller positive dimension. This yields a finite irreducible decomposition.

Export E2/v1. $V = W \oplus \ker P$ as representations, and every such V is a finite direct sum of irreducibles.

Acceptance gate. The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

Targeted repair. If repeated splitting is asserted to terminate without a measure, use the strictly decreasing dimensions of the summands. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Average a projection

1. **N1.1.** Choose a vector-space complement by extending a basis of W, and let p be its projection. Each conjugate of p lands in W, since W is invariant under every $\rho(g)$. On $w \in W$, that conjugate sends w to itself. Thus the average P also maps into W and is the identity there.

2. **N1.2.** For $h \in G$, conjugating P by $\rho(h)$ replaces the index g by hg, a permutation of G. Therefore P commutes with $\rho(h)$. Its restriction to its image is the identity, so $P^2 = P$ and $\operatorname{im} P = W$.

B. N2 — Split and terminate

1. **N2.1.** Every v decomposes as $Pv + (v - Pv)$, with the first term in W and the second in $\ker P$. Their intersection is zero because $P|_W$ is the identity. If $u \in \ker P$, then $P\rho(g)u = \rho(g)Pu = 0$, so the complement is invariant.

2. **N2.2.** For $\dim V = 0$ the decomposition is the empty sum. A nonzero irreducible V is already one summand. Otherwise choose a proper nonzero invariant W, split it as above, and apply the induction hypothesis to both summands of strictly smaller positive dimension. This yields a finite irreducible decomposition.

□

V. WORKED ORCHESTRATION

Let $G = C_2$ act on k^2 by swapping coordinates, with $\operatorname{char} k \neq 2$. For $W = k(1,1)$, start with $p(x,y) = (x,x)$. Averaging its two conjugates gives $P(x,y) = ((x+y)/2, (x+y)/2)$. Its kernel is $k(1,-1)$, yielding the trivial and sign summands.

A. Boundary and hypothesis audit

Over $\mathbb{F}_p$, let a generator of C_p act by $T = I + N$ on a two-dimensional space, where $Ne_2 = e_1$ and $Ne_1 = 0$. Then $T^p = I$. The invariant line ke_1 has no invariant complement: any complementary line has a vector $e_2 + ae_1$, whose image differs by e_1 and cannot stay in that line. Here $|G| = 0$ in the field.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 candidate: the same average proves complete reducibility in every characteristic. **Validation:** the unipotent C_p example contradicts the conclusion and $1/|G|$ is undefined. **Repair:** retain the stated invertibility assumption; no averaging repair exists for that counterexample under the broader claim. Recheck the complement and recursive decomposition with the correct field contract.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use invariant complements to decompose finite-group representations before character calculations; this does not itself prove the corresponding claim for arbitrary Lie algebras.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.